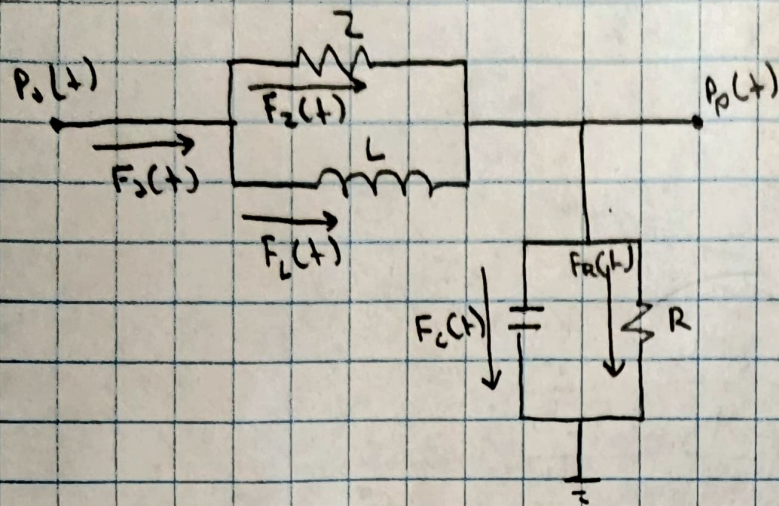


Práctica 5.4: Sistema cardiovascular



Ecuación principal

$$F_2(t) = F_z(t) + F_L(t) = F_L(t) + F_R(t)$$

$$F_z(t) = \frac{P_2(t) - P_p(t)}{Z}$$

$$F_C(t) = \frac{C dP_p(t)}{dt}$$

$$F_L(t) = \frac{1}{L} \int [P_2(t) - P_p(t)] dt$$

$$F_R = \frac{P_p(t)}{R}$$

Procedimiento algebraico

$$\frac{P_2(t)}{Z} - \frac{P_p(t)}{Z} + \frac{1}{L} \int [P_2(t) - P_p(t)] dt = \frac{C dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

Transformada de Laplace

$$\frac{P_2(s)}{Z} + \frac{P_p(s)}{Z} + \frac{P_2(s) - P_p(s)}{Ls} = CsP_p(s) + \frac{P_p(s)}{R}$$

$$s \left(\frac{1}{Z} + \frac{1}{Ls} \right) = \left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls} \right) P_p(s)$$

$$\frac{Ls + Z}{LsZ} P_2(s) = \left(\frac{CRLZs^2 + LZs + RLS + RZ}{RZLS} \right) P_p(s)$$

$$\frac{P_2(s)}{P_p(s)} = \frac{\frac{Ls + Z}{LsZ}}{\frac{CRLZs^2 + LZs + RLS + RZ}{RZLS}} = \frac{(Ls + Z)(RZLS)}{(LsZ)(CRLZs^2 + LZs + RLS + RZ)}$$

$$= \frac{RLS + RZ}{(CLRZs^2 + (LZ + RL)s + RZ)}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s P_2(s) \left[1 - \frac{P_p(s)}{P_2(s)} \right]$$

$$= \lim_{s \rightarrow 0} \cancel{s} \frac{1}{\cancel{s}} \left[1 - \frac{RLs + Rz}{(CLRz s^2 + (Lz + RL)s + Rz)} \right]$$

$$= 1 - \frac{Rz}{Rz} = 0V$$

Estabilidad en lazo abierto

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = CLRz$$

$$c = Rz$$

$$b = Lz + RL$$

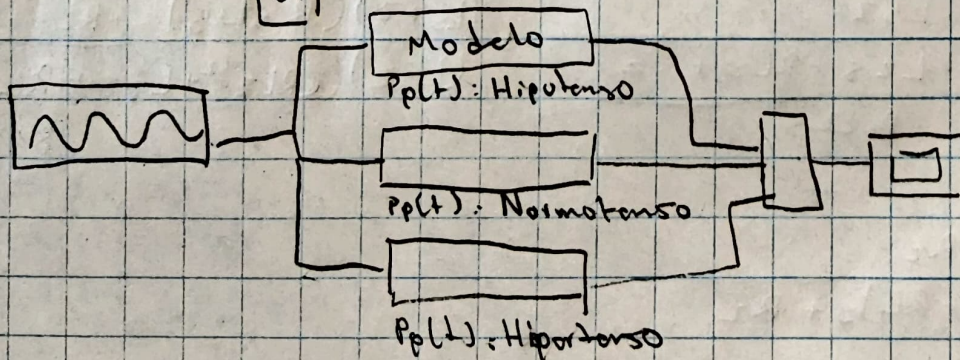
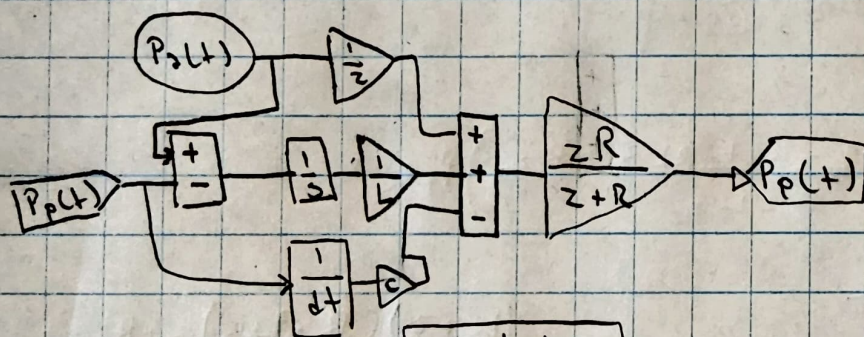
$$\lambda_{1,2} = \frac{-(Lz + RL) \pm \sqrt{(Lz + RL)^2 - 4CLR^2z^2}}{2CLRz} = \frac{- (?) }{+}$$

El sistema tiene una respuesta estable $\text{Re } \lambda_{1,2} < 0$

Modelo de ecuaciones integro-diferenciales

$$P_p(t) \left(\frac{1}{R} + \frac{1}{z} \right) = \frac{P_s(t)}{z} + \frac{1}{L} \int [P_s(t) - P_p(t)] dt + \frac{dP_p(t)}{dt}$$

$$P_p(t) = \left(\frac{P_s(t)}{z} + \frac{1}{L} \int [P_s(t) - P_p(t)] dt - \frac{dP_p(t)}{dt} \right) \frac{zR}{z+R}$$



$$\min = -0.2$$

$$\max = 1$$

$$\text{seed} = 106$$